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Solder wire as a spool (left) and in a tube (right)

Soldering

While it is possible to dabble in the world of electronics without really picking up a soldering iron, you'll discover a whole new world has opened up for you if you learn this simple skill. In this section, we will discuss how to choose the right solder for your needs, soldering irons and other accessories.

Picking the right solder

Solder is an alloy that usually presents as a long, thin wire in spools or tubes. Soldering is the process of joining two pieces of metal together in what is known as a solder joint. With manufacturing electronics, it is usually best to use lead-free solder to ensure the safety of your products. However, some people still prefer utilising the leaded solder because of its ability to act as a joining agent. It's your choice, whether you want to go for safety or functionality.

Lead-free solder essentially takes away some of the properties of solder which was what made it ideal for its intended purpose which is joining two pieces of metal. Another property is the melting point. Tin has a higher melting point than lead, so more heat is needed to achieve flow. Many lead-free solders have a flux core which is a chemical agent that aids the flowing of lead-free solder. You can use lead-free solder without flux, but it does make it easier to get the same effects as a leaded solder. Additionally, lead-free solder is also more expensive than the leaded variants.

There are also a number of other factors to consider. Firstly, there are numerous solder compositions out there apart from lead and tin which you can easily find by looking at the Wiki page of solder. Solder also comes in a variety of gauges or widths. Typically, when you are working with small components, you should use a very thin piece of solder. For larger components, stick to thicker wires. Additionally, solder also comes in forms other than wires. When you are getting into surface-mount soldering, solder paste is usually the recommended choice.